Federico Maguire

AI Trainer (RLHF) & Data Analyst · Python · SQL · Data Visualization

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SUMMARY

AI trainer and data analyst with RLHF experience at Scale AI, where I evaluated and stress-tested LLM responses to Python and SQL coding tasks across 10+ projects, including reviewer roles. Currently building FlowLayerPRO, a real-time analytics desktop app developed with an AI coding assistant under strict, evidence-based review of every generated change. Freelance work since 2023, including a CRM and a WhatsApp chatbot for clients; data analysis with Python, SQL, Excel, Tableau, and Power BI. Previously ran a CNC/CAD-CAM design business for 13 years, which brought hands-on G-code programming and a precision-first approach to technical work. English C2 (EF SET), native Spanish.

EXPERIENCE

Founder & Developer | FlowLayerPRO

Sep 2025 – Present

Remote · Personal product, in investor demos

- Designed and built a real-time order flow analytics desktop app for crypto futures traders, streaming live trades and order-book depth from Binance, Bybit, and OKX across 8 perpetual markets (Python engine, Tauri shell, Plotly.js WebGL frontend).
- Directed development with an AI coding assistant across 40+ documented milestones, reviewing every generated change against log or test evidence. Caught flawed designs, repeated misdiagnoses, false claims of completed work, and a verification command that leaked a credential.
- Resolved processing slowdowns of up to 2.9 s per cycle with a heap-based redesign of large-trade tracking and cut the refresh interval from 500 to 300 ms; validated through live-market stress tests of up to 4 h 41 min across all three exchanges with zero slow cycles.
- Diagnosed a WebSocket connection leak (3 → 21 open sockets) and thread race conditions using socket monitoring, then verified the fixes on the packaged release build.
- Deployed and operated a self-hosted licensing and payment server: hardened Ubuntu VPS, 6 Docker Compose services, Ed25519-signed licenses, HMAC-verified webhooks, and nightly encrypted backups with a tested restore. Set up GitHub Actions CI for Windows installer builds.
- Maintained a decision log and technical documentation. The production Windows build is complete and being shown in investor demos ahead of a closed beta.

AI Trainer – Python & SQL (RLHF) | Scale AI

May 2024 – Jun 2025

Remote · Contract

- Worked on 10+ RLHF and SFT projects, evaluating LLM responses to Python and SQL coding prompts for correctness, efficiency, safety, and adherence to best practices.
- Wrote adversarial coding prompts designed to make models fail within set time limits; difficulty increased as models improved across project stages.
- Created prompts for supervised fine-tuning (SFT), rated and compared model outputs, and evaluated code documentation quality.
- Served as a reviewer on several projects: audited contributor submissions, attended daily reviewer meetings, and supported contributors live with their tasks.
- Worked in project teams under project managers, adapting quickly to each project's guidelines and team structure.

Freelance Developer & Data Analyst | Maguire Data Analytics

Oct 2023 – Present

Remote · Freelance

- **CARMAGNO (Jul 2026 – Present):** consultant and collaborator on a multi-agency CRM for a car dealership owner, with an insurance-quoting module, live at carmagno.com.ar (PHP, MySQL, JavaScript; 85-table database).
- Translated the owner's business needs into specs and instructions for an AI coding agent: per-agency data isolation with a shared inventory, role-based access for 5 user types, and commission rules.

- Oversaw the AI's work, reviewing its output and steering it through fixes until each feature worked as the owner intended.
- **WhatsApp chatbot (2023):** built a customer-service chatbot for an Argentine business in Python (Flask, OpenAI API, WhatsApp Cloud API), with business-specific instructions and conversation storage in MySQL for later analysis.

Founder & Lead CAD/CAM Designer | **CNCstudio** Jan 2010 – Dec 2022

Buenos Aires, Argentina

- Founded and ran a CNC design and cutting business for 13 years, serving custom furniture clients, furniture makers, and boat builders.
- Turned client images and sketches into finished furniture: 3D modeling and rendering in Rhino, toolpath and G-code programming in Vectric Aspire and ArtCAM, and fabrication.
- Designed and produced product lines for 7 own and client brands (Housestyle, Lampmakers, Wamshop.ar, Objetos del Hogar, Dog's Life, Cajas Varias, MiniCarteles): furniture, lamps, home decor, pet products, packaging, and LED signage, covering design, production, assembly, and product photography.
- Digitized structural plywood templates and provided precision cutting for boat manufacturers, including Lanchas Bermuda; this became the business's main line of work.
- Led the design, programming, and construction of a custom 4th-axis tangential CNC cutter for specialized plastics.

SELECTED PROJECTS

- **2023 Data Job Postings Analysis** (PostgreSQL, Tableau): cleaned and queried 780,000 rows of salary data and built a Tableau world-map dashboard of salary distribution across data roles.
- **London Bike Sharing Analysis & Prediction** (Python, Tableau): EDA of seasonal and weather trends, an interactive Tableau dashboard, and a linear regression model to forecast rental demand.
- **Also in portfolio:** Python trading bot with live data collection and GUI, HR analysis (SQL + Power BI), coffee sales dashboard (Excel), radio marketing regression model (Python).

Portfolio: fedemaguire.github.io/MaguireDataAnalytics

SKILLS

AI Training: RLHF, SFT, LLM evaluation, adversarial prompting, code review and debugging, prompt engineering

Python: pandas, NumPy, scikit-learn, Matplotlib, Plotly/Dash, PyQt6, websocket-client, Requests; multithreading (threading, queue, heapq); packaging (Cython, PyInstaller)

Data & BI: SQL (PostgreSQL), Excel, Tableau, Power BI

Tools: Git, GitHub Actions, Docker Compose, Linux server administration

3D Design & CAD/CAM: Rhino 3D (modeling and rendering), Vectric Aspire, ArtCAM, G-code, Mach3, CNC, 3D printing, Arduino

CERTIFICATIONS & COURSES

- **Google Advanced Data Analytics Professional Certificate**, Google / Coursera (Oct 2023)
- **Preprocessing Unstructured Data for LLM Applications**, DeepLearning.AI (May 2024)
- **ChatGPT Prompt Engineering for Developers**, DeepLearning.AI / OpenAI (May 2023)
- **Python for Machine Learning & Data Science Masterclass**, Udemy (Jun 2023)
- **Complete Python, SQL, and Excel Bootcamps**, Pierian Data / Udemy (Mar–May 2023)
- **Computational Thinking for Problem Solving**, University of Pennsylvania / Coursera (Apr 2023)
- **Career Essentials in Data Analysis**, Microsoft / LinkedIn (Apr 2023)
- **Python for Finance and Algorithmic Trading with QuantConnect**, Udemy (Aug 2024)

LANGUAGES

Spanish: Native **English:** C2 Proficient (EF SET 77/100, Mar 2024)